

• Naïve Bayes recommendatⁿ sys.

- are type of collaborative filtering technique used in recommendatⁿ sys.
- The idea behind this is to recommend an item to a user based on the preferences of similar users.

How it works ??????

1] User-Item Matrix :-

The matrix is maintained by system where rows represents the users and columns represents the items and cells have ratings. However, not all cells are filled, since user typically rate only a small subset of items.

2] User similarity Calcⁿ :-

The sys calc similarity betⁿ the users based on their rating patterns. by using metrics like cosine similarity, pearson correlatⁿ coeff^t etc.

3] Neighborhood Selection:-
for target users, system identifies subset of similar users, called as Neighborhood

4] Recommendation Generation:-
Once the Nei. is determined, sys identifies items that the target user has not rated but have been ~~see~~ highly rated by users in neighborhood. These items are recommended

* 5] Rating Aggregation:-

The system may also aggregate ratings from neighborhood to provide predicted rating for each recommended item.

∴ Limitations:-

Sparsity:-

means most cells are empty because users typically rate only a small fraction of items.

∴ 2) Cold start problem :-

New users or items may not have enough rating to accurately determine their similarity with other users.

3) Scalability :-

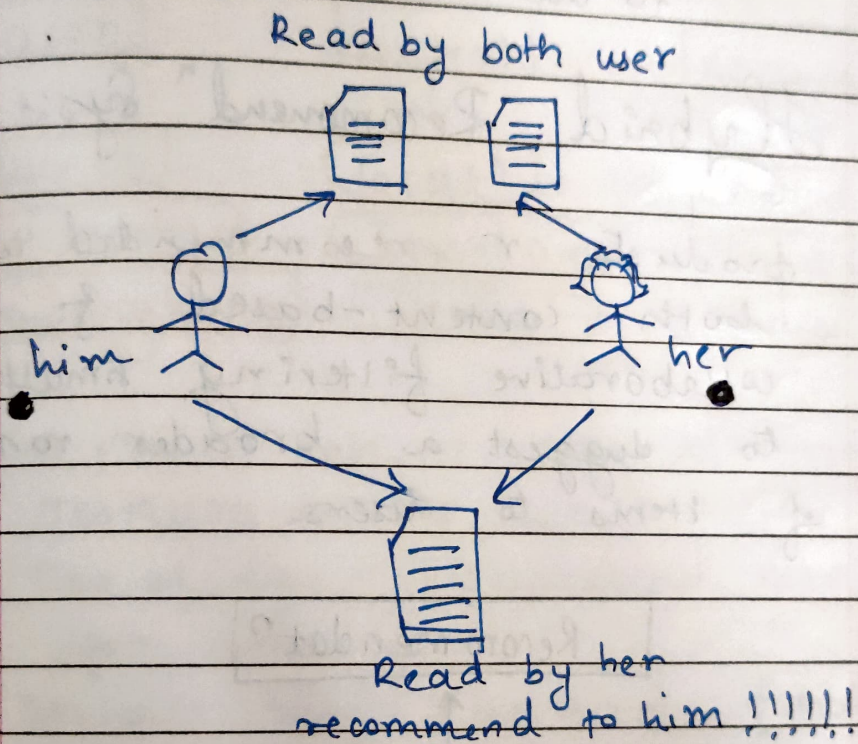
As no. of user & items grow, calcⁿ user similarity becomes computationally expensive.

4) Limited Serendipity :-

Limits system's ability to introduce new & diverse items.

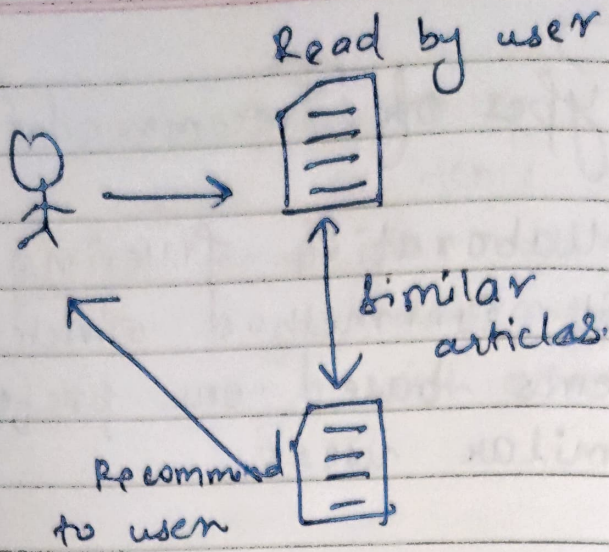
Types of Recommendation System

1. Collaborative filtering:-
It is method which recommends items based on preference of similar users.



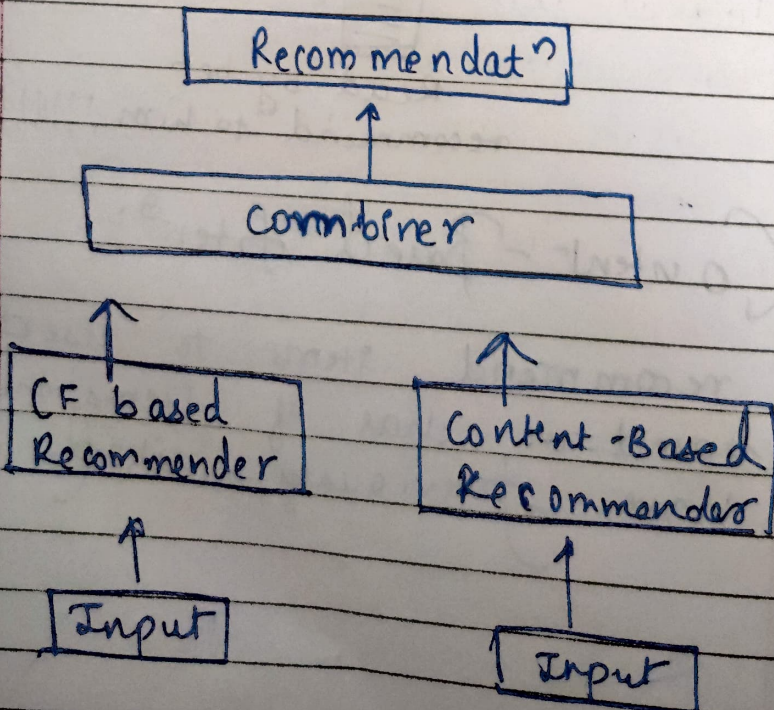
2. Content-Based filter:-

recommend items to users based on char of items they have previously liked.



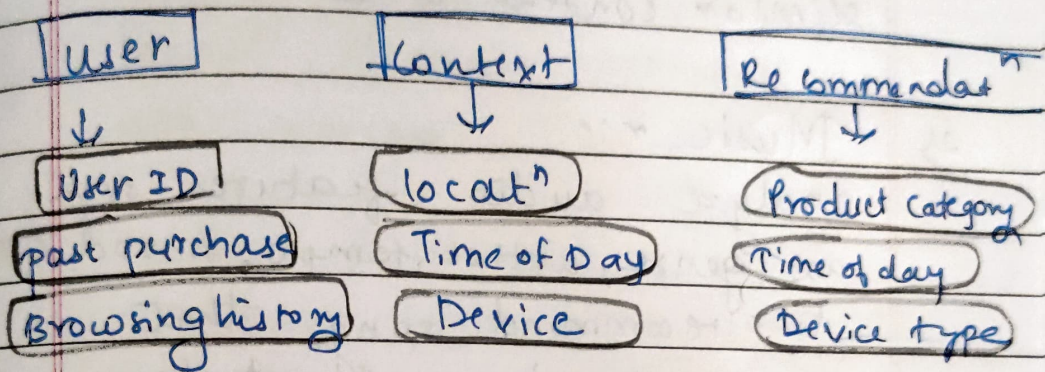
3. Hybrid Recommendation Sys:-

products r recommended using both content-based & collaborative filtering simultaneously to suggest a broader range of items to users.



4) Context - Aware

- consider additional contextual inf, such as user locatⁿ, time, device to tailor recommendⁿ to user's current situatⁿ.



→ In product cate, it may recommend restaurants for lunch etc

Time of day - recommend movies for evening

Device - may recommend if mobile friendly product

Applicatⁿ → Content Based R.S.

1) Movie & TV show R:-

can analyze attributes of movie, such as genre, actors, directors, user rating, to recommend similar content to user

2) Music R:-

analyze audio features such as genre, artist, tempo, mood, to recommend songs that match user's music taste.

3) News & Article R:-

can analyze textual content of news articles or blog posts, topics, writing style, to rec-

4) Book Recomⁿ:-

genre, author, publicatⁿ year, plot summary

5) Recipe R:-

analyze ingredients, cuisine type, cooking method, user ratings.

- 6) Educational content r:-
analyze edu resources such as
subject area, difficulty, learning
objectives, etc.
- 7) Travel Recommendation:-
travel destination, such as
location, climate, attractions,
user reviews.
- 8) Health & fitness.
exercise type, duration, intensity,
user goals, meal plans, etc.
- 9) Job R:-
such as job title, industry,
location, req. skills, experience, etc.
- 10) Product R:-
e-commerce platforms --
prod. category, brand, price

Alternat⁹ least sq:-

- Collaborative filter⁹ recommend items to user based on preferences of similar users.
- A matrix is created where rows → user, columns → items, & cells have ratings. It is often sparse, with many missing entries.
- Matrix factorizatⁿ aims to decompose their user-item matrix into ~~two larger~~ two lower dimensional matrices representatⁿ latent factors.

ALS for Optimizatⁿ:-

- 1) Initializatⁿ:-
Start with randomly initialized ~~to~~ matrices for user factors (U) and item factors (V).
- 2) Alternat⁹ Optimizatⁿ
 - a) Fix items factors.

- 2b. • For each user, calc predicted preferences for all items based on current user ~~fixed~~ factors (U) & fixed item factor (V).
- 2c. • Use LEAST SQUARE minimization to adjust user factors (U) to minimize diff betⁿ predicted preferences & actual user preferences.
- Fix user factors & repeat steps 2b-2c, but this time optimize item factor (V) based on fixed user factor (U).

3) Repeat:-
Continue iterating betⁿ steps 2b-2c until a stopping criterion is met.

AI	ML	DL:
→ simulates human intelligence an to perform tasks and make decisions.	→ It is subset of AI & learns from the data patterns from data using alg.	→ It is subset of ML which has artificial neural networks
→ It may or may not require large datasets it can use predefined rules	→ It heavily relies on labeled data set for training data	→ It requires extensive labeled data, performs good on complex datasets
→ It is rule-based, so requires human intervent ⁿ & programming.	→ It req. less manual intervent ⁿ	→ It req. less manual engineering
→ can handle various tasks, from simple to complex.	→ It performs tasks like classification, regression, etc	→ excels tasks like img recognit ⁿ , NLP, etc
→ AI is used in virtual assistants, recommendat ⁿ sys.	→ MLs used in spam filtering, img recognit ⁿ , & data tasks	→ uses in autonomous vehicle, speech recog, etc.

Training

- 1] The ML model learns from the training data.
- 2] The model gains knowledge from training data to make accurate predictions.
- 3] The training data distribution is similar to actual data distribution.
- 4] To stop overfitting, training data is utilized.
- 5] Size is typically larger.

Testing

- 1] The testing data is used to assess the model's performance.
- 2] The testing data is not exposed to model until evaluation.
- 3] The testing data distribution is different from the real world distribution.
- 4] By making predictions on testing data & comparing them to actual labels, the model's performance is assessed.
- 5] Smaller.

Good

Accuracy	Precision	Recall	F1-Score
proportion of correctly classified instances.	proportion of true +ve predictions	proportion of true +ve predicted out of actual positive (+ve)	harmonic mean of precision & recall
$\frac{\text{no. of correct prediction}}{\text{Total no. of prediction}}$	$\frac{TP}{TP + FP}$	$\frac{TP}{TP + FN}$	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
Application - General classifier performance assessment.	- emphasizes minimizing False Positive	" False neg	- harmonic mean of P & R
Eg. NLP, Sentiment Analysis -	Eg. Medical diagnosis	fraud detect ⁿ	Information retrieval systems